Student Admission:

The dataset simulates a university admission record system, where each student’s admission profile includes test scores, high school percentages, and admission status. The data contains realistic flaws often encountered in raw data, offering hands-on experience in data wrangling.

Target: Apply Dataset Exploration and Preprocessing and then Predict Admission Status column.

Diabetes Dataset:

The diabetes_prediction_dataset.csv file contains medical and demographic data of patients along with their diabetes status, whether positive or negative. It consists of various features such as age, gender, body mass index (BMI), hypertension, heart disease, smoking history, HbA1c level, and blood glucose level. The Dataset can be utilized to construct machine learning models that can predict the likelihood of diabetes in patients based on their medical history and demographic details.

Target: Apply Dataset Exploration and Preprocessing and then Predict whether he is diabetic or not.

Marketing Campaign:

Customer Personality Analysis is a detailed analysis of a company’s ideal customers. It helps a business to better understand its customers and makes it easier for them to modify products according to the specific needs, behaviors and concerns of different types of customers. Customer personality analysis helps a business to modify its product based on its target customers from different types of customer segments. For example, instead of spending money to market a new product to every customer in the company’s database, a company can analyze which customer segment is most likely to buy the product and then market the product only on that particular segment.

Target: Apply Dataset Exploration and Preprocessing and then Predict the response of the customer.

Employee Dataset:

This dataset can be used for various HR and workforce-related analyses, including employee retention, salary structure assessments, diversity and inclusion studies, and leave pattern analyses. Researchers, data analysts, and HR professionals can gain valuable insights from this dataset.

- Potential Research Questions:
- What is the distribution of educational qualifications among employees?
- How does the length of service (Joining Year) vary across different cities?
- Is there a correlation between Payment Tier and Experience in Current Domain?
- What is the gender distribution within the workforce?
- Are there any patterns in leave-taking behavior among employees?

Target: Apply Dataset Exploration and Preprocessing and then Predict whether the employee is going to leave or not.

Customer Segmentation Dataset:

This case requires to develop a customer segmentation to define marketing strategy. The sample Dataset summarizes the usage behavior of about 9000 active credit card holders during the last 6 months. The file is at a customer level with 18 behavioral variables. Apply Clustering for customer segmentation.

Following is the Data Dictionary for Credit Card dataset :-

- CUST_ID : Identification of Credit Card holder (Categorical)
- BALANCE : Balance amount left in their account to make purchases (
- BALANCE_FREQUENCY : How frequently the Balance is updated, score between 0 and 1 (1 = frequently updated, 0 = not frequently updated)
- PURCHASES : Amount of purchases made from account
- ONEOFF_PURCHASES : Maximum purchase amount done in one-go
- INSTALLMENTS_PURCHASES : Amount of purchase done in installment
- CASH_ADVANCE : Cash in advance given by the user
- PURCHASES_FREQUENCY : How frequently the Purchases are being made, score between 0 and 1 (1 = frequently purchased, 0 = not frequently purchased)
- ONEOFFPURCHASESFREQUENCY : How frequently Purchases are happening in one-go (1 = frequently purchased, 0 = not frequently purchased)
- PURCHASESINSTALLMENTSFREQUENCY : How frequently purchases in installments are being done (1 = frequently done, 0 = not frequently done)
- CASHADVANCEFREQUENCY : How frequently the cash in advance being paid
- CASHADVANCETRX : Number of Transactions made with "Cash in Advanced"
- PURCHASES_TRX : Numbe of purchase transactions made
- CREDIT_LIMIT : Limit of Credit Card for user
- PAYMENTS : Amount of Payment done by user
- MINIMUM_PAYMENTS : Minimum amount of payments made by user
- PRCFULLPAYMENT : Percent of full payment paid by user
- TENURE : Tenure of credit card service for user

Fraud Detection Dataset:

Target: Apply Dataset Exploration and Preprocessing and then Predict whether the transaction is fraud or not.

Heart Disease Dataset:

Target: Apply Dataset Exploration and Preprocessing and then Predict whether the person has heart disease or not.

Booking Dataset:

Target: Apply Dataset Exploration and Preprocessing and then predict whether a reservation will be confirmed or not based on the booking attributes provided.

Mall Customers Dataset:

Target: Apply Dataset Exploration and Preprocessing and then group mall customers into distinct segments based on their demographic and spending behavior to better understand their purchasing patterns.

Insurance Dataset:

Target: Apply Dataset Exploration and Preprocessing and then predict how do factors such as age, sex, BMI, number of children, smoking status, and region influence insurance charges.

Housing Price Dataset:

Target: Apply Dataset Exploration and Preprocessing and then predict the price of a house based on its features such as area, number of bedrooms and bathrooms, availability of amenities, and furnishing status.

الفرقة	السكاشن	Datasets
الفرقة الرابعه	سكشن ١	1. Student Admission. 2. Mall Customers. 3. Insurance Dataset.
	سكشن ٢	1. Diabetes Dataset. 2. Marketing Campaign. 3. Housing Price Dataset.
	سكشن ٣	1. Employee Dataset. 2. Customer Segmentation Dataset 3. Insurance Dataset.
	سكشن ٤	1. Fraud Detection Dataset. 2. Heart Disease Dataset. 3. Housing Price Dataset.
	سكشن ٥	1. Booking Dataset. 2. Mall Customers Dataset. 3. Housing Price Dataset.
	سكشن ٦	1. Student Admission Dataset. 2. Booking Dataset. 3. Insurance Dataset.
	سكشن ٧	1. Employee Dataset. 2. Mall Customer Dataset.
الفرقة الثالثة	سكشن ١	1. Diabetes Dataset. 2. Marketing Campaign. 3. Housing Price Dataset.
	سكشن ٢	1. Fraud Detection Dataset. 2. Heart Disease Dataset. 3. Housing Price Dataset.
	سكشن ٣	1. Student Admission Dataset. 2. Booking Dataset. 3. Insurance Dataset.
	سكشن ٤	1. Employee Dataset. 2. Customer Segmentation Dataset 3. Insurance Dataset.